

The Complexity Trap: Simple Observation Masking Is as Efficient as LLM Summarization for Agent Context Management

Tobias Lindenbauer*, Igor Slinko, Ludwig Felder, Egor Bogomolov, Yaroslav Zharov

Simple observation masking matches the raw agent solve rate at 50% lower cost.

Motivation

Practical:

- LLM agents are a critical test-time scaling technique for solving complex problems.
- Their iterative approach results in fast context growth resulting in exploding costs.

Underexplored:

- Related work typically focuses on improving the effectiveness or solve rate with little consideration for efficiency concerns.

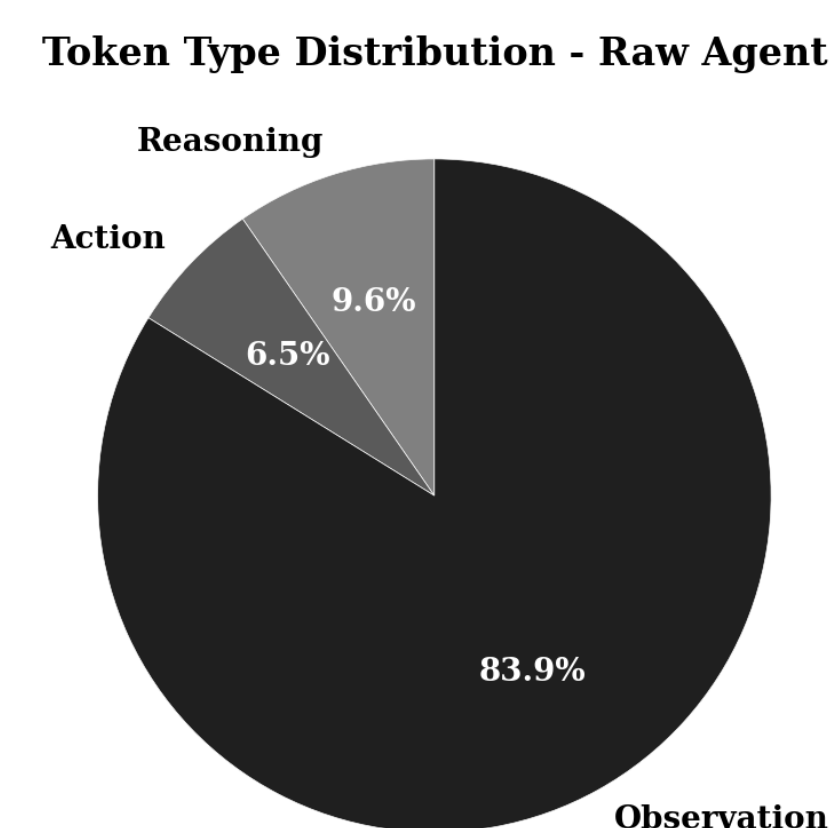


Figure: Environment observation tokens dominate the context window of an SE agent's trajectory.

Context Management Strategies

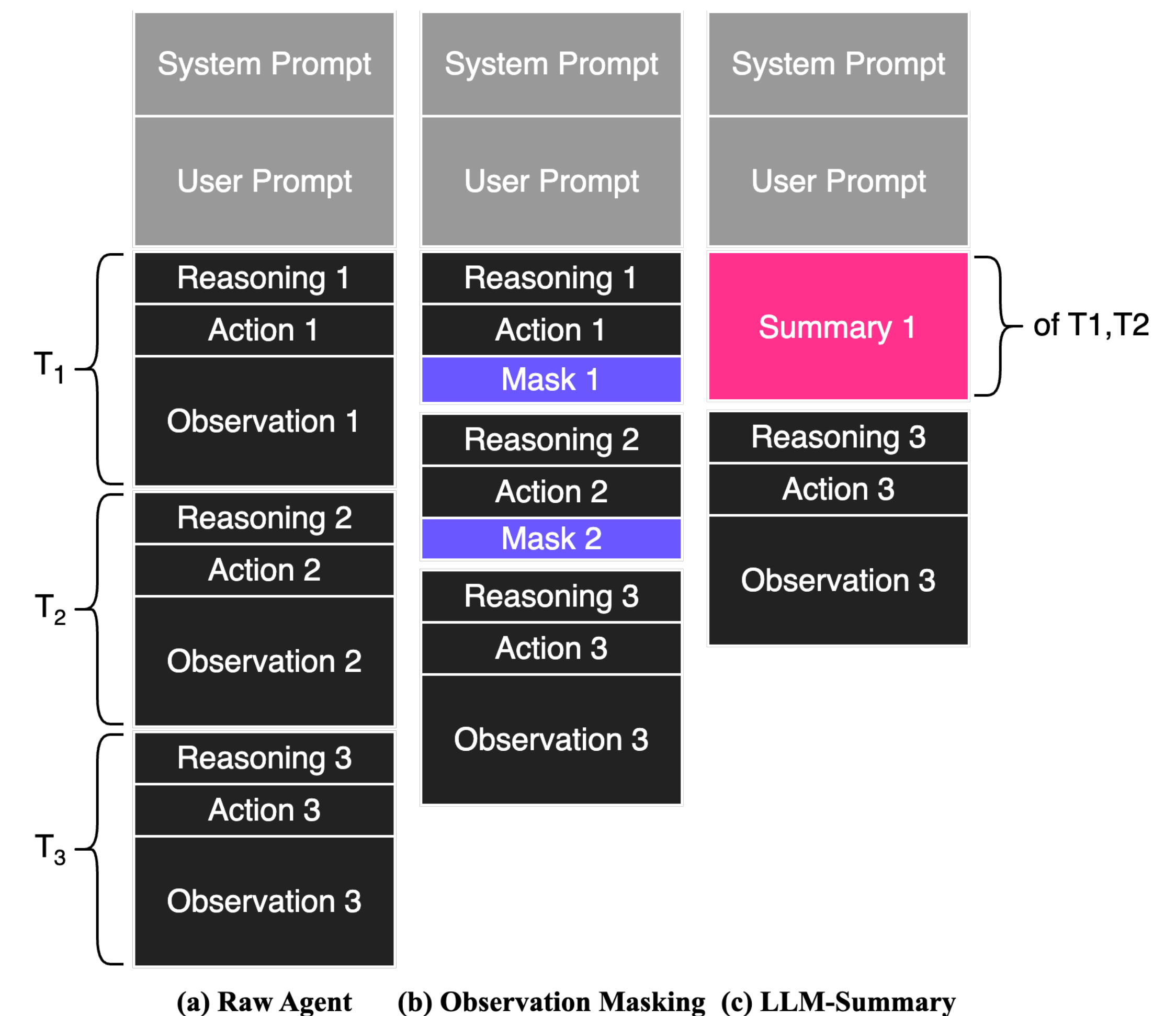


Figure: Box heights indicate the number of tokens in that portion of a typical trajectory. (a) The baseline ReAct-style [4] trajectory; context grows with each action-observation pair. (b) **LLM-based summarization** condenses older turns into a running summary and preserving a few recent turns in full (e.g., OpenHands [2]). (c) **Observation masking** replaces observations older than a fixed window of M turns (here, $M=1$) with a placeholder (e.g., SWE-agent [3])

Main Results

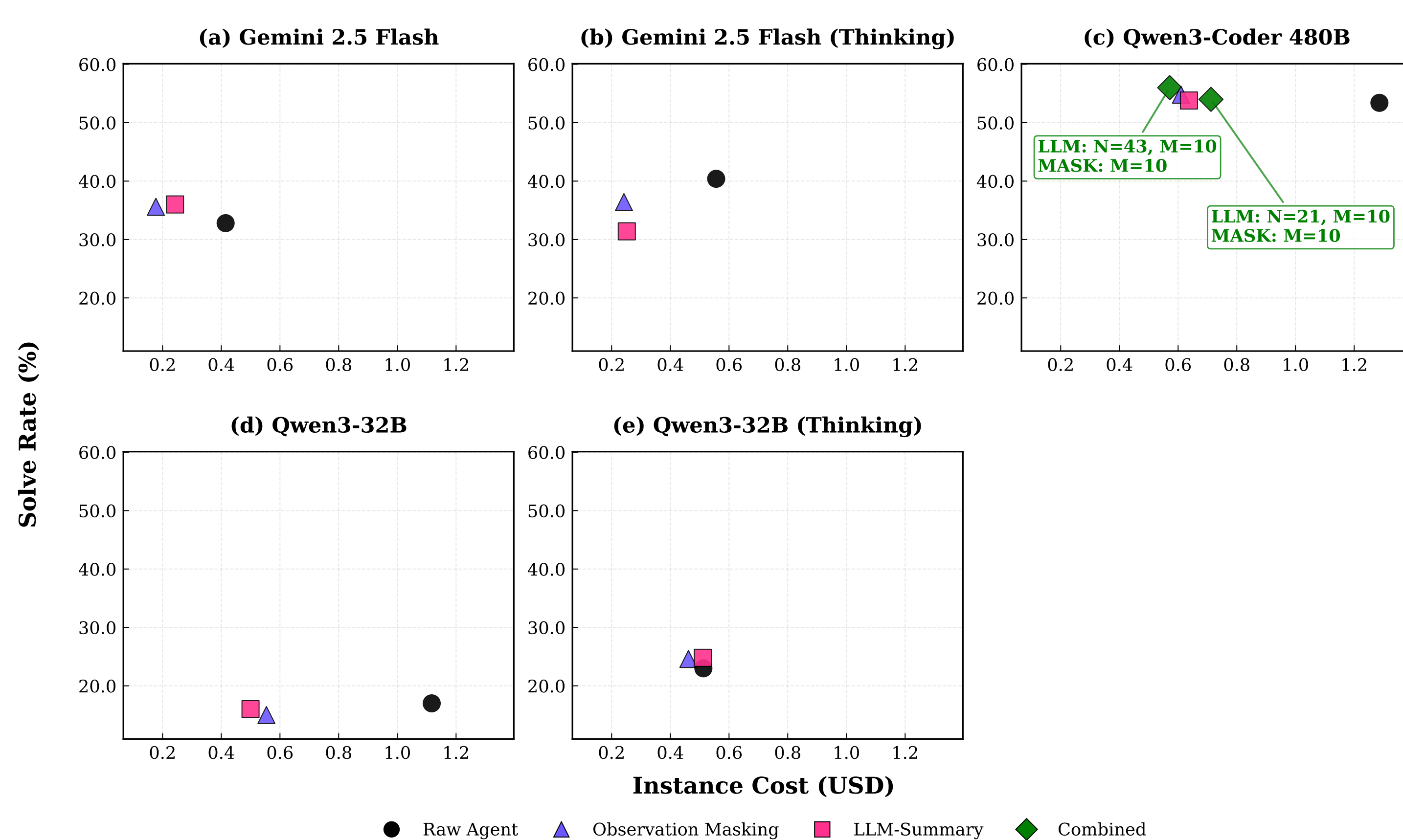


Figure: The effectiveness versus efficiency tradeoff for context management strategies within SWE-agent [3] on SWE-bench Verified [1]. The plot compares solve rate (y-axis, $\uparrow$) against the average cost per trajectory (x-axis, $\downarrow$) for different model configurations. Across all models, the Observation Masking strategy consistently occupies the most efficient frontier, achieving solve rates competitive with, and sometimes superior to, the LLM-Summary strategy. Given the right hyperparameters, combining the two strategies further improves solve rate and reduces costs.

Trajectory Elongation

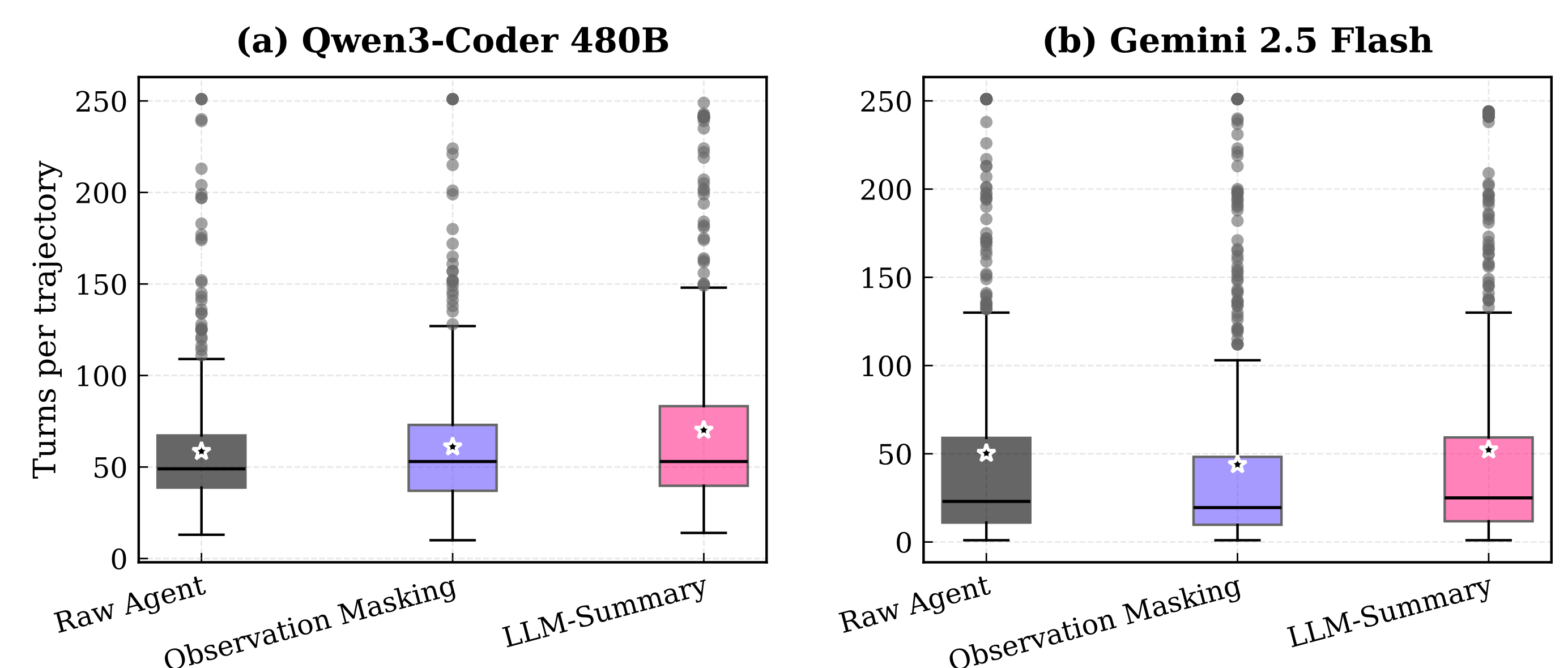


Figure: Impact of context management strategies on trajectory length. Box plots show the distribution of trajectory lengths (in turns) across different strategies within SWE-agent [3]. LLM-Summary consistently leads to longer trajectories, suggesting they mask failure signals that would otherwise prompt earlier termination. The star indicates the mean trajectory length.

Ablation: Agent Scaffold

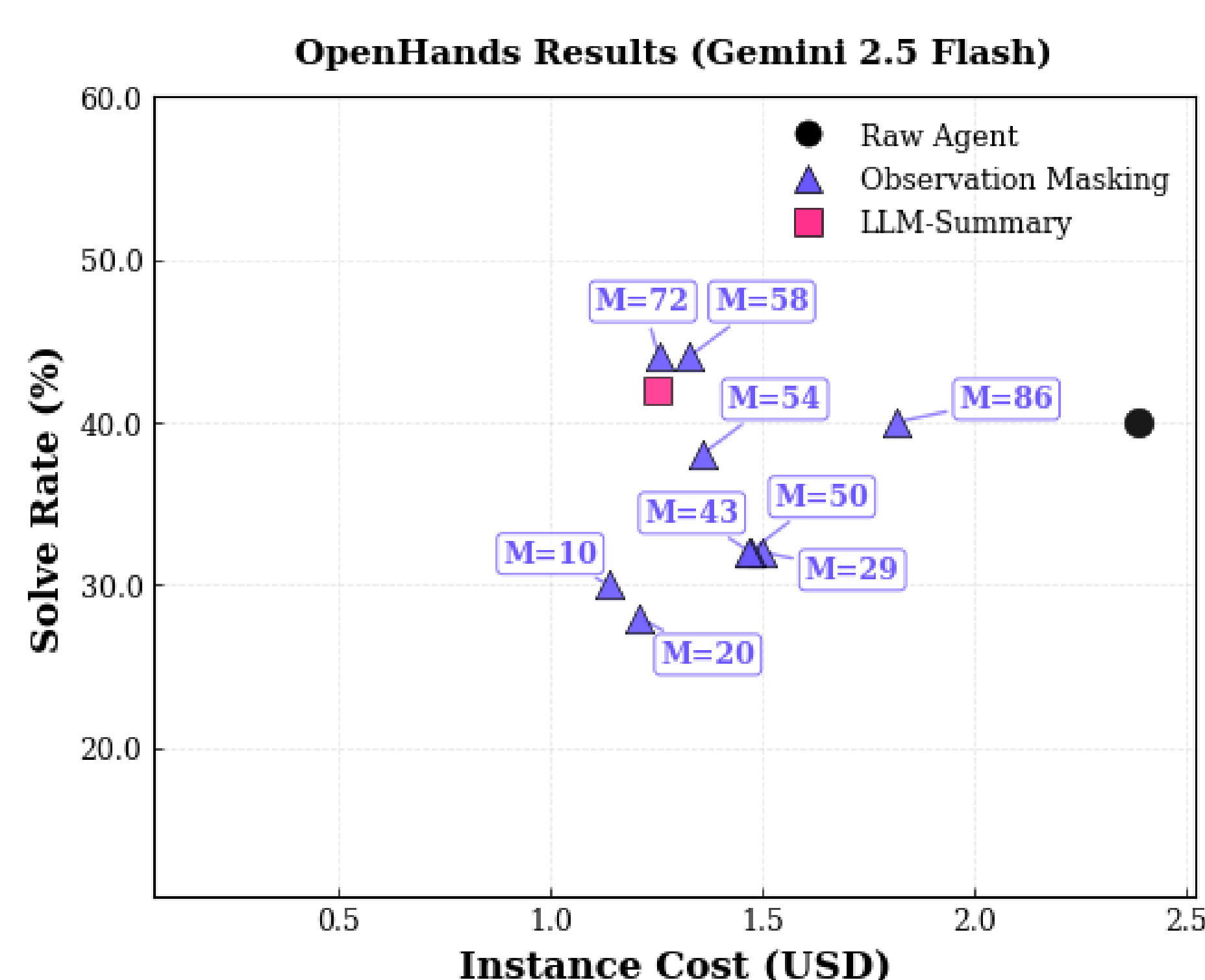


Figure: Observation Masking exhibits a similarly strong performance on OpenHands [2] given the right hyperparameter value for the rolling window size M . This provides evidence for the generality of our findings concerning agent scaffolds.

Links



Read the paper



Build on our findings



Contact the authors

Bibliography

- [1] Neil Chowdhury et al. *Introducing SWE-bench Verified*. 2024.
- [2] Xingyao Wang et al. "OpenHands: An Open Platform for AI Software Developers as Generalist Agents".
- [3] John Yang et al. "SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering".
- [4] Shunyu Yao et al. "ReAct: Synergizing Reasoning and Acting in Language Models".